



EPS.MR: R package/ Python library to compute quantiles, perform local-regression and construct KD-trees in parallel.

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Abstract

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1. Transformations with Chebyshev's Polynomials

In mathematics the Chebyshev's polynomials, named after Pafnuty Chebyshev, are a sequence of orthogonal polynomials which can be defined recursively. The Chebyshev's polynomials of the first kind are defined by the recurrence relation $T_0(t) = 1$; $T_1(t) = t$; and $T_{j+1}(t) = 2t \cdot T_j(t) - T_{j-1}(t)$ for $j > 0$ and $t \in \mathbb{R}$. The expressions of first few non-trivial Chebyshev's polynomials for $t \in \mathbb{R}$ are as follows,

$$\begin{aligned}T_2(t) &= 2t^2 - 1 \\T_3(t) &= 4t^3 - 3t \\T_4(t) &= 8t^4 - 8t^2 + 1 \\T_5(t) &= 16t^5 - 20t^3 + 5t\end{aligned}$$

The expression of a general j^{th} order Chebyshev's polynomial can be obtained recursively as,

$$T_j(t) = \frac{j}{2} \sum_{j'=0}^{\left\lfloor \frac{j}{2} \right\rfloor} (-1)^{j'} \frac{(j-j'-1)!}{j'!(j-j')!} (2t)^{j-2j'}, \text{ for } j \in \mathbb{N}^+ \text{ and } t \in \mathbb{R}.$$

From this expression, we realize an odd-degree Chebyshev's polynomial $T_{2j-1}(t)$ is a non-zero linear combination of $t, t^3, \dots, t^{2j-1}$ for $t \in \mathbb{R}$, and an even-degree Chebyshev's polynomial

$T_{2j}(t)$ is a non-zero linear combination of $1, t^2, t^4, \dots, t^{2j}$ for $t \in \mathbb{R}$. Now for $j \in \mathbb{N}^+$ and $z \in (0, 1)$, we have the following trigonometric identities:

$$\begin{aligned} T_{2j-1}(\cos(z)) &= \cos((2j-1)z) \\ T_{2j-1}(\sin(z)) &= (-1)^{j-1} \sin((2j-1)z). \end{aligned}$$

1.1. 1-dimensional transformations with Chebyshev's Polynomials

Let $c_{2j-1}(z) = \cos((2j-1)z)$ and $c_{2j}(z) = \sin((2j-1)z)$ for $j \in \mathbb{N}^+$, also let, $c^{2j-1}(z) = \cos^{2j-1}(z)$ and $c^{2j}(z) = \sin(z) \cos^{2j-2}(z)$ for $j \in \mathbb{N}^+$. Observe that, $c_1(z) = c^1(z) = \cos(z)$ and $c_2(z) = c^2(z) = \sin(z)$. For $J \in \mathbb{N}^+$, let us define vector valued functions:

$$\begin{aligned} c_{(1:2J)}(z) &= (c_1(z), c_2(z), \dots, c_{2J}(z)), \text{ and} \\ c^{(1:2J)}(z) &= (c^1(z), c^2(z), \dots, c^{2J}(z)). \end{aligned}$$

Also, for consistency in future results, for $J \in \mathbb{N}^+$ we define $\tilde{c}_{(1:2J)}(z, J) = c_{(1:2J)}(z)$ and $\tilde{c}^{(1:2J)}(z, J) = c^{(1:2J)}(z)$. Now, we have the following lemma:

Lemma 1 Given $J \in \mathbb{N}^+$, $\exists$ linear transformation $\tau_J^{(A)} : \mathbb{R}^{2J} \rightarrow \mathbb{R}^{2J}$, such that:

$$c_{(1:2J)}(z) = \tau_J^{(A)}(c^{(1:2J)}(z)).$$

Proof 1 First, remember we have the following Chebyshev's identities for Cosines:

$$c_{2j-1}(z) = \cos((2j-1)z) = T_{2j-1}(\cos(z)).$$

Here $T_{2j-1}(t)$ is the $(2j-1)^{th}$ degree Chebyshev's polynomial, and we have the following expression of $T_{2j-1}(t)$ for $j \in \mathbb{N}^+$:

$$T_{2j-1}(t) = \frac{(2j-1)}{2} \sum_{j'=1}^j (-1)^{j-j'} \frac{(j+j'-2)!}{(j-j')!(2j'-1)!} (2t)^{2j'-1}$$

If we replace t with $\cos(z)$, we get for $j \in \mathbb{N}^+$:

$$\begin{aligned} c_{2j-1}(z) &= \cos((2j-1)z) = T_{2j-1}(\cos(z)) \\ &= \sum_{j'=1}^j (-1)^{j-j'} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \cos^{2j'-1}(z) \\ &= \sum_{j'=1}^j (-1)^{j-j'} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \cdot c^{2j'-1}(z). \end{aligned}$$

For $j \in \{1, 2, \dots, J\}$, define:

$$\begin{aligned} a_{2j-1, 2j'-1} &= (-1)^{j-j'} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \text{ for } 1 \leq j' \leq j, \\ a_{2j-1, 2j'-1} &= 0 \text{ for } j < j' \leq J \text{ and} \\ a_{2j-1, 2j'} &= 0 \text{ for } 1 \leq j' \leq J. \end{aligned}$$

Then, from the expression of $c_{2j-1}(z)$, we can say that:

$$c_{2j-1}(z) = \sum_{j'=1}^{2J} a_{2j-1, j'} \cdot c^{j'}(z) \text{ for } 1 \leq j \leq J.$$

Now remember, we also have the Chebyshev's identities for Sines:

$$c_{2j}(z) = \sin((2j-1) \cdot z) = (-1)^{j-1} T_{2j-1}(\sin(z))$$

In the previous expression of $T_{2j-1}(z)$, if we replace x with $\sin(z)$, we get for $j \in \mathbb{N}^+$:

$$\begin{aligned} c_{2j}(z) &= \sin((2j-1) \cdot z) = (-1)^{j-1} T_{2j-1}(\sin(z)) \\ &= \sum_{j'=1}^j (-1)^{j'-1} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \sin^{2j'-1}(z). \end{aligned}$$

Now, observe that $\sin^{2j'-1}(z) = \sin(z) - \sum_{j''=1}^{j'-1} \sin(z) \cos^{2j''}(z)$ for $j' > 1$, then for $j > 1$, we have:

$$\begin{aligned} c_{2j}(z) &= \sin((2j-1) \cdot z) = (-1)^{j-1} T_{2j-1}(\sin(z)) \\ &= (2j-1) \sin(z) + \sum_{j'=2}^j (-1)^{j'-1} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \sin^{2j'-1}(z) \\ &= (2j-1) \sin(z) \\ &\quad + \sum_{j'=2}^j (-1)^{j'-1} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \left(\sin(z) - \sum_{j''=1}^{j'-1} \sin(z) \cos^{2j''}(z) \right) \\ &= \left(\sum_{j'=1}^j (-1)^{j'-1} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \right) \sin(z) \\ &\quad + \sum_{j'=2}^j \sum_{j''=1}^{j'-1} \left((-1)^{j'} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \sin(z) \cos^{2j''}(z) \right) \\ &= \left(\sum_{j''=1}^j (-1)^{j''-1} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \right) \sin(z) \\ &\quad + \sum_{j'=2}^j \left(\sum_{j''=j'}^j (-1)^{j''} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \right) \sin(z) \cos^{2j'-2}(z) \\ &= \left(\sum_{j''=1}^j (-1)^{j''-1} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \right) \cdot c^2(z) \\ &\quad + \sum_{j'=2}^j \left(\sum_{j''=j'}^j (-1)^{j''} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \right) \cdot c^{2j'}(z). \end{aligned}$$

Given $j \in \{1, 2, \dots, J\}$, we define:

$$\begin{aligned} a_{2j, 2j'-1} &= 0 \text{ for } 1 \leq j' \leq J, \\ a_{2j, 2} &= \sum_{j''=1}^j (-1)^{j''-1} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \text{ and} \\ a_{2j, 2j'} &= \sum_{j''=j'}^j (-1)^{j''} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \text{ for } 2 \leq j' \leq j, \\ a_{2j, 2j'} &= 0 \text{ for } j+1 \leq j' \leq J. \end{aligned}$$

Then, from the expression of $c_{2j}(z)$, we can also say that:

$$c_{2j}(z) = \sum_{j'=1}^{2J} a_{2j, j'} \cdot c^{j'}(z) \text{ for } 1 \leq j \leq J.$$

Hence, in general we can say:

$$c_j(z) = \sum_{j'=1}^{2J} a_{j, j'} \cdot c^{j'}(z) \text{ for } 1 \leq j \leq 2J.$$

Define the $2J \times 2J$ matrix $\mathcal{A}_J^{(A)} = \{\{a_{j, j'}\}\}_{1 \leq j \leq 2J, 1 \leq j' \leq 2J}$. Then, we can say that:

$$c_{(1:2J)}(z) = \mathcal{A}_J^{(A)} \cdot c^{(1:2J)}(z).$$

So, we proved existence of a linear transformation $\tau_J^{(A)} : \mathbb{R}^{2J} \rightarrow \mathbb{R}^{2J}$, such that:

$$c_{(1:2J)}(z) = \tau_J^{(A)}(c^{(1:2J)}(z)).$$

with the explicit representation $\tau_J^{(A)}(\mathbf{v}) = \mathcal{A}_J^{(A)} \cdot \mathbf{v}$.

Remark: Form the alternate definition we have:

$$\tilde{c}_{(1:2J)}(z, J) = \tau_J^{(A)}(\tilde{c}^{(1:2J)}(z, J)).$$

1.2. 2-dimensional transformations with Chebyshev's Polynomials

Let us introduce another set of functions for $j \in \mathbb{N}^+$, given $x \in \mathbb{R}$ and $J' \in \mathbb{N}^+$: a fixed positive integer.

$$\begin{aligned} \dot{c}_j(z, J') &= \cos(2 \cdot (j-1) \cdot J' \cdot z) = \cos((j-1) \cdot (2J'z)), \\ \dot{c}^j(z, J') &= \cos^{j-1}(2J'z). \end{aligned}$$

Also, let us define vector valued functions for $J \in \mathbb{N}^+$:

$$\begin{aligned} \dot{c}_{(1:J)}(z, J') &= (\dot{c}_1(z, J'), \dot{c}_2(z, J'), \dots, \dot{c}_J(z, J')), \\ \dot{c}^{(1:J)}(z, J') &= (\dot{c}^1(z, J'), \dot{c}^2(z, J'), \dots, \dot{c}^J(z, J')). \end{aligned}$$

Then, we have the following lemma:

Lemma 2 Given a fixed $J' \in \mathbb{N}^+$, for $J \in \mathbb{N}^+$, there exist linear transformation $\tau_J^{(B)} : \mathbb{R}^J \rightarrow \mathbb{R}^J$, such that:

$$\dot{c}_{(1:J)}(z, J') = \tau_J^{(B)}(\dot{c}^{(1:J)}(z, J')).$$

Proof 2 First given any $J' \in \mathbb{N}^+$, observe $\dot{c}_0(z, J') = 1$, $\dot{c}_1(z, J') = \cos(2J'z)$ and $\dot{c}^0(z, J') = 1$, $\dot{c}^1(z, J') = \cos(2J'z)$ for any $x \in \mathbb{R}$. For $j > 2$, we again use the Chebyshev's identities. For a fixed $J' \in \mathbb{N}^+$, we have for $x \in \mathbb{R}$:

$$\dot{c}_j(z, J') = \cos((j-1) \cdot (2J'z)) = T_{j-1}(\cos(2J'z)).$$

Here $T_j(z)$ is the j^{th} degree Chebyshev's polynomial, and remember we have the following expression of $T_j(z)$ for $j > 1$,

$$T_j(t) = \frac{j}{2} \sum_{k=0}^{\lfloor \frac{j}{2} \rfloor} (-1)^k \frac{(j-k-1)!}{k!(j-k)!} (2t)^{j-2k}.$$

If we replace t with $\cos(2J'z)$ and j with $j-1$, we get for $j \in \mathbb{N}^+$:

$$\begin{aligned} \dot{c}_j(z, J') &= \cos((j-1) \cdot (2J'z)) = T_{j-1}(\cos(2J'z)) \\ &= \sum_{k=0}^{\lfloor \frac{j-1}{2} \rfloor} (-1)^k \frac{2^{j-2k-2} j(j-k-2)!}{k!(j-k-1)!} \cos^{j-2k-1}(2J'z) \\ &= \sum_{k=0}^{\lfloor \frac{j-1}{2} \rfloor} \left((-1)^k \frac{2^{j-2k-2} j(j-k-2)!}{k!(j-k-1)!} \right) \cdot \dot{c}^{j-2k}(z, J'). \end{aligned}$$

Let us define:

$$\begin{aligned} a_{1,1} &= 1, \quad a_{1,j} = 0 \text{ for } 2 \leq j \leq J; \\ a_{2,1} &= 0, \quad a_{2,2} = 1, \quad a_{2,j} = 0 \text{ for } 2 < j \leq J; \\ a_{j,k} &= (-1)^k \frac{2^{j-2k-2} j(j-k-2)!}{k!(j-k-1)!} \text{ if } j \geq 2, \quad k \leq j, \quad \frac{j-k-1}{2} \in \mathbb{N}^+; \\ a_{j,k} &= 0 \text{ if } j \geq 2, \quad k \leq j, \quad \frac{j-k-1}{2} \notin \mathbb{N}^+ \text{ or } j < k \leq J. \end{aligned}$$

Then, from the expression of $\dot{c}_j(z, J')$, we can say that:

$$\dot{c}_j(z, J') = \sum_{k=1}^J a_{j,k} \cdot \dot{c}^k(z, J') \text{ for } 1 \leq j \leq J.$$

Define $J \times J$ matrix $\mathcal{A}_J^{(B)} = \{\{a_{j,k}\}\}_{1 \leq j \leq J, 1 \leq k \leq J}$. Then, we can say:

$$\dot{c}_J(z, J') = \mathcal{A}_J^{(B)} \cdot \dot{c}^J(z, J').$$

So, we proved existence of a linear transformation $\tau_J^{(B)} : \mathbb{R}^J \rightarrow \mathbb{R}^J$, such that:

$$\dot{c}_J(z, J') = \tau_J^{(B)}(\dot{c}^J(z, J')).$$

with the explicit representation $\tau_J^{(B)}(\mathbf{x}) = \mathcal{A}_J^{(B)} \cdot \mathbf{x}$.

1.3. Transformations on matrices

Consider two integers $J_1 \in \mathbb{N}^+$ and $J_2 \in \mathbb{N}^+$. Define:

$$\mathbb{N}_{\{J_1, J_2\}} := \{1, \dots, 2J_1\} \times \{1, \dots, J_2\}.$$

Define 2 dimensional sequential functions:

$$\begin{aligned}\tilde{c}_{j_1, j_2}(z, J_1, J_2) &= c_{j_1}(z) \cdot \dot{c}_{j_2}(z, J_1). \\ \tilde{c}^{j_1, j_2}(z, J_1, J_2) &= c^{j_1}(z) \cdot \dot{c}_{j_2}(z, J_1).\end{aligned}$$

Also, define matrix-valued functions:

$$\begin{aligned}\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2) &= \{\{\tilde{c}_{j_1, j_2}(z, J_1, J_2)\}\}_{(j_1, j_2) \in \mathbb{N}_{\{J_1, J_2\}}}. \\ \tilde{c}^{(1:2J_1), (1:J_2)}(z, J_1, J_2) &= \{\{\tilde{c}^{j_1, j_2}(z, J_1, J_2)\}\}_{(j_1, j_2) \in \mathbb{N}_{\{J_1, J_2\}}}.\end{aligned}$$

Lemma 3 *Given integers $J_1 \in \mathbb{N}^+$ and $J_2 \in \mathbb{N}^+$, there exist linear transformation $\tau_{J_1, J_2}^{A, B} : \mathbb{R}^{2J_1} \times \mathbb{R}^{J_2} \rightarrow \mathbb{R}^{2J_1} \times \mathbb{R}^{J_2}$, such that:*

$$\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2) = \tau_{J_1, J_2}^{A, B}(\tilde{c}^{(1:2J_1), (1:J_2)}(z, J_1, J_2)).$$

Proof 3 *Let us consider the j_2^{th} column of the matrix $\tilde{c}^{(1:2J_1), (1:J_2)}(z, J_1, J_2)$ which is the array:*

$$\begin{aligned}\tilde{c}^{(1:2J_1), j_2}(z, J_1, J_2) &= (\tilde{c}^{1, j_2}(z, J_1, J_2), \dots, \tilde{c}^{2J_1, j_2}(z, J_1, J_2)) \\ &= (c^1(z) \cdot \dot{c}^{j_2}(z, J_1), \dots, c^{2J_1}(z) \cdot \dot{c}^{j_2}(z, J_1)) \\ &= (c^1(z), \dots, c^{2J_1}(z)) \cdot \dot{c}^{j_2}(z, J_1) = c^{(1:2J_1)}(z) \cdot \dot{c}^{j_2}(z, J_1).\end{aligned}$$

So, $\tilde{c}^{(1:2J_1), j_2}(z, J_1, J_2)$ is a scalar multiple of the array $c^{(1:2J_1)}(z)$ with multiplier $\dot{c}^{j_2}(z, J_1)$. Then, for the transformation $\tau_J^{(A)}$ in `chev:lem:trA` (with $J = J_1$), we have:

$$\begin{aligned}\tau_{J_1}^{(A)}(\tilde{c}^{(1:2J_1), j_2}(z, J_1, J_2)) &= \tau_{J_1}^{(A)}(c^{(1:2J_1)}(z) \cdot \dot{c}^{j_2}(z, J_1)) \\ &= \tau_{J_1}^{(A)}(c^{(1:2J_1)}(z)) \cdot \dot{c}^{j_2}(z, J_1) = c_{(1:2J_1)}(z) \cdot \dot{c}^{j_2}(z, J_1) \\ &= \tilde{c}_{(1:2J_1), 0}^{0, j_2}(z, J_1, J_2) = (\tilde{c}_{1, 0}^{0, j_2}(z, J_1, J_2), \dots, \tilde{c}_{2J_1, 0}^{0, j_2}(z, J_1, J_2)); \\ \text{when: } \tilde{c}_{j_1, 0}^{0, j_2}(z, J_1, J_2) &:= c_{j_1}(z) \cdot \dot{c}^{j_2}(z, J_1), \text{ for } j_1 \in \{1, 2, \dots, 2J_1\}.\end{aligned}$$

Hence we realize, if we sequentially apply the linear transformation $\tau_{J_1}^{(A)}$ along each column of the matrix $\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$, we get the matrix-valued statistic:

$$\tilde{c}_{(1:2J_1), 0}^{0, (1:J_2)}(z, J_1, J_2) = \{\{\tilde{c}_{j_1, 0}^{0, j_2}(z, J_1, J_2)\}\}_{(j_1, j_2) \in \mathbb{N}_{\{J_1, J_2\}}}.$$

Now, let us look at the j_1^{th} row of the matrix $\tilde{c}_{(1:2J_1), 0}^{0, (1:J_2)}(z, J_1, J_2)$, which is the array:

$$\begin{aligned}\tilde{c}_{j_1, 0}^{0, (1:J_2)}(z, J_1, J_2) &= (c_{j_1, 0}^{0, 1}(z, J_1, J_2), \dots, c_{j_1, 0}^{0, J_2}(z, J_1, J_2)) \\ &= (c_{j_1}(z) \cdot \tilde{c}^1(z, J_1), \dots, c_{j_1}(z) \cdot \tilde{c}^1(z, J_1)) \\ &= c_{j_1}(z) \cdot (\tilde{c}^1(z, J_1), \dots, \tilde{c}^1(z, J_1)) = c_{j_1}(z) \cdot \tilde{c}^{(1:J_2)}(z, J_1).\end{aligned}$$

Again, we realize $\tilde{c}_{j_1, 0}^{(1:J_2)}(z, J_1, J_2)$ is a scalar-multiple of the array $\tilde{c}^{(1:J_2)}(z, J_1)$ with multiplier $c_{j_1}(z)$. Now, using the transformation $\tau_j^{(B)}$ in `chev:lem:trB` (with $J = J_2$), we conclude:

$$\begin{aligned} \tau_{J_2}^{(B)}(\tilde{c}^{(1:2J_1), j_2}(z, J_1, J_2)) &= \tau_{J_2}^{(B)}(c_{j_1}(z) \cdot \dot{c}^{(1:J_2)}(z, J_1)) \\ &= c_{j_1}(z) \cdot \tau_{J_2}^{(B)}(\dot{c}^{(1:J_2)}(z, J_1)) = c_{j_1}(z) \cdot \dot{c}^{(1:J_2)}(z, J_1) \\ &= \tilde{c}_{j_1, (1:J_2)}(z, J_1, J_2) = (\tilde{c}_{j_1, 1}(z, J_1, J_2), \dots, \tilde{c}_{j_1, J_2}(z, J_1, J_2)); \\ \text{when: } \tilde{c}_{j_1, j_2}(z, J_1, J_2) &:= c_{j_1}(z) \cdot \dot{c}_{j_2}(z, J_1), \text{ for } j_2 \in \{1, 2, \dots, J_2\}. \end{aligned}$$

So if we sequentially apply the linear transformation $\tau_{J_2}^{(B)}$ along each row of the matrix $\tilde{c}_{(1:2J_1), 0}^{(1:J_2)}(z, J_1, J_2)$, we'll get with the matrix $\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$. So we have demonstrated a sequence of linear transformation that transforms the matrix $\tilde{c}^{(1:2J_1), (1:J_2)}(z, J_1, J_2)$ into the matrix $\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$. In other words we established the existence of a linear transformation $\tau_{J_1, J_2}^{(A, B)}$, such that:

$$\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2) = \tau_{J_1, J_2}^{(A, B)}(\tilde{c}^{(1:2J_1), (1:J_2)}(z, J_1, J_2)).$$

Now, observe that for $J_1 \in \mathbb{N}^+$ and $J_2 \in \mathbb{N}^+$:

$$c_{(1:2J_1 J_2)}(z) = (c_1(z), \dots, c_{2J_1 J_2}(z, J_1, J_2)), \text{ we have the following result:}$$

Lemma 4 Given integers $J_1 \in \mathbb{N}^+$ and $J_2 \in \mathbb{N}^+$, there exist a linear-transformation $\tau_{J_1, J_2}^{(C)} : \mathbb{R}^{2J_1} \times \mathbb{R}^{J_2} \rightarrow \mathbb{R}^{2J_1 J_2}$, such that:

$$c_{(1:2J_1 J_2)}(z) = \tau_{J_1, J_2}^{(C)}(\tilde{c}_{(1:2J_1), (1:J_2)}(z)) \text{ for } x \in (0, 1).$$

Proof 4 First, let us adopt to the following notation for $j_2 \in \{1, \dots, J_2\}$:

$$c_{2J_1(j_2-1)+(1:2J_2)}(z) = (c_{2J_1(j_2-1)+1}, \dots, c_{2J_1(j_2-1)+2J_1}).$$

Note that, if we concatenate: $c_{2J_1(1-1)+(1:2J_2)}(z), \dots, c_{2J_1(J_2-1)+(1:2J_2)}(z)$ in a single array, we'll get $c_{(1:2J_1 J_2)}(z)$. Now, we are going to prove our proposed statement: $c_{2J_1(j_2-1)+(1:2J_2)}(z)$ can be obtained from the matrix $\tilde{c}_{(1:2J_1), (1:J_2)}(z)$ by linear-transformations for $j_2 \in \{1, 2, \dots, J_2\}$, by principles of mathematical induction. For $j_2 = 1$, note that:

$$c_{2J_1(1-1)+(1:2J_2)}(z) = (c_1(z), \dots, c_{2J_1}(z)) = c_{(1:2J_1)}(z).$$

Also note that the first column of $\tilde{c}_{2J_1, J_2}(z)$ is:

$$\begin{aligned} &(\tilde{c}_{1, 1}(z, J_1, J_2), \dots, \tilde{c}_{2J_1, 1}(z, J_1, J_2)) \\ &= (c_1(z) \cdot \dot{c}_1(z, J_1), \dots, c_{2J_1}(z) \cdot \dot{c}_1(z, J_1)) \\ &= (c_1(z), \dots, c_{2J_1}(z)) = c_{(1:2J_1)}(z). \text{ (since } \dot{c}_1(z, J_1) = 1 \text{).} \end{aligned}$$

So our proposed statement is trivially true. Now we will assume that for some $j_1 \in \{2, 3, \dots, J_2-1\}$ our proposed statement is true, and we will prove it for $j_1 + 1$. By our assumption each element in the array:

$$c_{2J_1(j_2-1)+(1:2J_2)}(z) = (c_{2J_1(j_2-1)+1}, \dots, c_{2J_1(j_2-1)+2J_1}).$$

can be obtained from $\tilde{c}_{(1:2J_1), (1:J_2)}(z)$ by linear-transformations. Note that for any $x \in (0, 1)$, $\{J_1, J_2\} \in \mathbb{N}^+$ and $j_1 \in \{1, 2, \dots, J_1\}$, $j_2 \in \{1, 2, \dots, J_2\}$, we have the following trigonometric identities:

$$\begin{aligned} & \cos\left((2J_1j_2 + (2j_1 - 1))z\right) + \cos\left((2J_1j_2 - (2j_1 - 1))z\right) \\ &= 2 \cos(2J_1j_2z) \cos((2j_1 - 1)z), \\ & \sin\left((2J_1j_2 + (2j_1 - 1))z\right) - \sin\left((2J_1j_2 - (2j_1 - 1))z\right) \\ &= 2 \cos(2J_1j_2z) \sin((2j_1 - 1)z). \end{aligned}$$

With little rearrangements, can rewrite these two equations as,

$$\begin{aligned} c_{2J_1j_2+2j_1-1}(z) &= 2\tilde{c}_{2j_1-1, j_2}(z, J_1, J_2) - c_{2J_1j_2-2j_1+1}(z). \\ c_{2J_1j_2+2j_1}(z) &= 2\tilde{c}_{2j_1, j_2}(z, J_1, J_2) + c_{2J_1j_2-2j_1+2}(z). \end{aligned}$$

Observe that, if $j_1 \in \{1, 2, \dots, J_1\}$, then $(2J_1-2j_1+1) \in \{1, 3, \dots, 2J_1-1\}$ and $(2J_1-2j_1+2) \in \{2, 4, \dots, 2J_1\}$. So, by assumption, both $c_{2J_1j_2-2j_1+1}(z)$ and $c_{2J_1j_2-2j_1+2}(z)$ can be obtained from the matrix $\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$ by linear-transformations for $j_1 \in \{1, 2, \dots, J_1\}$. Then, from the relations mentioned above, we can say that, both $c_{2J_1j_2+2j_1-1}$ and $c_{2J_1j_2+2j_1}$ can be obtained from the matrix $\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$ by linear-transformations for $j_1 \in \{1, 2, \dots, J_1\}$. Hence, the array $c_{2J_1j_2+(1:2J_1)}(z)$ can be obtained from the matrix $\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$ by linear-transformations. So, we proved our induction hypothesis for $j_2 + 1$.

This conclusion proves our induction hypothesis, $c_{2J_1(j_2-1)+(1:2J_1)}(z)$ can be obtained from $\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$ by linear-transformations for $j_2 \in \{1, 2, \dots, J_2\}$, and hence the concatenated $2J_1J_2$ element array $c_{(1:2J_1J_2)}(z)$ can be obtained from the matrix $\tilde{c}_{(1:2J_1), (1:J_2)}(z)$ by linear transformations. So, we have proved existence of a linear transformation $\tau_J^{(C)} : \mathbb{R}^{2J_1} \times \mathbb{R}^{J_2} \rightarrow \mathbb{R}^{2J_1J_2}$, such that:

$$c_{(1:2J_1J_2)}(z) = \tau_{J_1, J_2}^{(C)}(\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)).$$

1.4. K -dimensional generalizations

First we would like to lay out few notations that we are going to use frequently in this section. For a general $K \in \mathbb{N}^+$ and $K > 2$, consider integer vector: $\underline{\mathbf{J}} = (J_1, J_2, \dots, J_K) \in \mathbb{N}^{+K}$. Define integers: $J := \prod_{k=1}^K J_k$ and $L_k := \prod_{l=1}^k J_l$ for $1 \leq k < K$. Also define the index set:

$$\mathbb{N}_{\{\underline{\mathbf{J}}\}} := \{1, \dots, 2J_1\} \times \{1, \dots, J_2\} \times \dots \times \{1, \dots, J_K\}.$$

For $\underline{\mathbf{j}} = (j_1, j_2, \dots, j_K) \in \mathbb{N}_{\{\underline{\mathbf{J}}\}}$, define K dimensional sequential functions:

$$\begin{aligned} \tilde{c}_{\underline{\mathbf{j}}}(z, \underline{\mathbf{J}}) &= \tilde{c}_{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}}) := c_{j_1}(z) \cdot \prod_{k=2}^K \dot{c}_{j_k}(z, L_{k-1}), \\ \tilde{c}^{\underline{\mathbf{j}}}(z, \underline{\mathbf{J}}) &= \tilde{c}^{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}}) := c^{j_1}(z) \cdot \prod_{k=2}^K \dot{c}^{j_k}(z, L_{k-1}). \end{aligned}$$

We also define K -dimensional array-valued functions:

$$\begin{aligned}\tilde{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}}) &= \{\{\tilde{c}_{\mathbf{j}}(z, \underline{\mathbf{J}})\}\}_{\{\mathbf{j} \in \mathbb{N}_{(\underline{\mathbf{J}})}\}}, \\ \tilde{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}}) &= \{\{\tilde{c}^{\mathbf{j}}(z, \underline{\mathbf{J}})\}\}_{\{\mathbf{j} \in \mathbb{N}_{(\underline{\mathbf{J}})}\}}.\end{aligned}$$

Then, we have the following result:

Lemma 5 *Given $\underline{\mathbf{J}} \in \mathbb{N}^{+K}$, $\exists$ a linear transformation*

$$\begin{aligned}\tau_{\underline{\mathbf{J}}}^{(K)} : \mathbb{R}^{2J_1} \times \mathbb{R}^{J_2} \times \dots \times \mathbb{R}^{J_K} &\rightarrow \mathbb{R}^{2J} \text{ such that:} \\ c_{(1:2J)}(z) &= \tau_{\underline{\mathbf{J}}}^{(K)}(\tilde{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})).\end{aligned}$$

Proof 5 *Let us fix the indices $j_2, \dots, j_K$ and traverse through the 1st dimension of the K -dimensional array $\tilde{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$. So, we change the first index j_1 over the values: $j_1 = 1, 2, \dots, 2J_1$, while keeping $j_2, \dots, j_K$ fixed and consider every-term $\tilde{c}^{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}})$ for such combinations of indices in $\mathbb{N}_{(\underline{\mathbf{J}})}$, which is the array:*

$$\begin{aligned}\tilde{c}_{(1:2J_1), j_2, \dots, j_K}(z, \underline{\mathbf{J}}) &= (\tilde{c}^{1, j_2, \dots, j_K}(z, \underline{\mathbf{J}}), \dots, \tilde{c}^{2J_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}})) \\ &= \left(c^1(z) \cdot \prod_{k=2}^K \tilde{c}^{j_k}(z, L_{k-1}), \dots, c^{2J_1}(z) \cdot \prod_{k=2}^K \tilde{c}^{j_k}(z, L_{k-1}) \right) \\ &= (c^1(z), \dots, c^{2J_1}(z)) \cdot \prod_{k=2}^K \tilde{c}^{j_k}(z, L_{k-1}) = c^{(1:2J_1)}(z) \cdot \prod_{k=2}^K \tilde{c}^{j_k}(z, L_{k-1}).\end{aligned}$$

So, $\tilde{c}_{(1:2J_1), j_2, \dots, j_K}(z, \underline{\mathbf{J}})$ is a scalar-multiple of $c^{(1:2J_1)}(z)$ with the multiplier $\prod_{k=2}^K \tilde{c}^{j_k}(z, L_{k-1})$. Then, for $\tau_{J_1}^{(A)}$ in *chev:lem:trA* (with $J = J_1$) we get:

$$\begin{aligned}\tau_{J_1}^{(A)}(\tilde{c}_{(1:2J_1), j_2, \dots, j_K}(z, \underline{\mathbf{J}})) &= \tau_{J_1}^{(A)}\left(c^{(1:2J_1)}(z) \cdot \prod_{k=2}^K \tilde{c}^{j_k}(z, L_{k-1})\right) \\ &= \tau_{J_1}^{(A)}(c^{(1:2J_1)}(z)) \cdot \prod_{k=2}^K \tilde{c}^{j_k}(z, L_{k-1}) = c_{(1:2J_1)}(z) \cdot \prod_{k=2}^K \tilde{c}^{j_k}(z, L_{k-1}) \\ &= \tilde{c}_{(1:2J_1), 0, \dots, 0}^{0, j_2, \dots, j_K}(z, \underline{\mathbf{J}}) = (\tilde{c}_{1, 0, \dots, 0}^{0, j_2, \dots, j_K}(z, \underline{\mathbf{J}}), \dots, \tilde{c}_{2J_1, 0, \dots, 0}^{0, j_2, \dots, j_K}(z, \underline{\mathbf{J}}));\end{aligned}$$

$$\text{where, } \tilde{c}_{j_1, 0, \dots, 0}^{0, j_2, \dots, j_K}(z, \underline{\mathbf{J}}) := c_{j_1}(z) \cdot \prod_{k=2}^K \tilde{c}^{j_k}(z, L_{k-1}),$$

for $j_1 \in \{1, 2, \dots, 2J_1\}$.

Hence, if we sequentially apply $\tau_{J_1}^{(A)}$ along 1st dimension of the K -dimensional array: $\tilde{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$, we'll get the K -dimensional array:

$$\tilde{c}_{(1:2J_1), 0, \dots, 0}^{0, (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}}) = \{\{\tilde{c}_{j_1, 0, \dots, 0}^{0, j_2, \dots, j_K}(z, \underline{\mathbf{J}})\}\}_{\{\mathbf{j} \in \mathbb{N}_{(\underline{\mathbf{J}})}\}}.$$

Next, we fix j_1 and $j_3, \dots, j_K$ and traverse through the 2^{nd} dimension of the K -dimensional array: $\tilde{c}_{(1:2J_1), \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}, \dots, \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}}^{0, (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$, for $j_2 \in \{1, 2, \dots, J_2\}$, which is the array:

$$\begin{aligned} & \tilde{c}_{j_1, \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}, \dots, \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}}^{0, (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}}) = (\tilde{c}_{j_1, 0, \dots, 0}^{0, 1, \dots, j_K}(z, \underline{\mathbf{J}}), \dots, \tilde{c}_{j_1, 0, \dots, 0}^{0, J_2, \dots, j_K}(z, \underline{\mathbf{J}})) \\ & = \left(c_{j_1}(z) \cdot \dot{c}^1(z, J_1) \cdot \prod_{k=3}^K \dot{c}^{j_k}(z, L_{k-1}), \dots, \right. \\ & \quad \left. c_{j_1}(z) \cdot \dot{c}^{J_2}(z, J_1) \cdot \prod_{k=3}^K \dot{c}^{j_k}(z, L_{k-1}) \right) \\ & = c_{j_1}(z) \cdot (\dot{c}^1(z, J_1), \dots, \dot{c}^{J_2}(z, J_1)) \cdot \prod_{k=3}^K \dot{c}^{j_k}(z, L_{k-1}) \\ & = c_{j_1}(z) \cdot \dot{c}^{(1:J_2)}(z, J_1) \cdot \prod_{k=3}^K \dot{c}^{j_k}(z, L_{k-1}). \end{aligned}$$

So, once again, we realize $\tilde{c}_{j_1, \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}, \dots, \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}}^{0, (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}})$ is a scalar-multiple of the array: $\dot{c}^{(1:J_2)}(z, J_1)$ with the multiplier $c_{j_1}(z) \cdot \prod_{k=3}^K \dot{c}^{j_k}(z, L_{k-1})$. In other words, this time, for $\tau_{J_2}^{(B)}$ in *chev:lem:trB* (with $J = J_2$) we get:

$$\begin{aligned} & \tau_{J_2}^{(B)}(\tilde{c}_{j_1, \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}, \dots, \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}}^{0, (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}})) \\ & = \tau_{J_2}^{(B)}\left(c_{j_1}(z) \cdot \dot{c}^{(1:J_2)}(z, J_1) \cdot \prod_{k=3}^K \dot{c}^{j_k}(z, L_{k-1})\right) \\ & = c_{j_1}(z) \cdot \tau_{J_2}^{(B)}\left(\dot{c}^{(1:J_2)}(z, J_1)\right) \cdot \prod_{k=3}^K \dot{c}^{j_k}(z, L_{k-1}) \\ & = c_{j_1}(z) \cdot \dot{c}^{(1:J_2)}(z, J_1) \cdot \prod_{k=3}^K \dot{c}^{j_k}(z, L_{k-1}) \\ & = \tilde{c}_{j_1, (1:2J_1), \dots, \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}}^{0, 0, \dots, j_K}(z, \underline{\mathbf{J}}) = (\tilde{c}_{j_1, 1, \dots, 0}^{0, 0, \dots, j_P}(z, \underline{\mathbf{J}}), \dots, \tilde{c}_{j_1, J_2, \dots, 0}^{0, 0, \dots, j_P}(z, \underline{\mathbf{J}})). \end{aligned}$$

$$\text{where, } \tilde{c}_{j_1, j_2, \dots, \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}}^{0, 0, \dots, j_P}(z, \underline{\mathbf{J}}) = c_{j_1}(z) \cdot \dot{c}^{j_2}(z, J_1) \cdot \prod_{k=3}^K \dot{c}^{j_k}(z, L_k),$$

for $j_2 \in \{1, 2, \dots, J_2\}$.

So, if we sequentially apply the linear transformation $\tau_{J_2}^{(B)}$ along the 2^{nd} dimension of the K -dimensional array: $\tilde{c}_{(1:2J_1), \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}, \dots, \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}}^{0, (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$, we will get the K -dimensional array:

$$\tilde{c}_{(1:2J_1), (1:J_2), \dots, \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}}^{0, 0, \dots, (1:J_K)}(z, \underline{\mathbf{J}}) = \{\{\tilde{c}_{j_1, j_2, \dots, 0}^{0, 0, \dots, j_P}(z, \underline{\mathbf{J}})\}\}_{\{\underline{\mathbf{j}} \in \mathbb{N}_{(\underline{\mathbf{J}})}\}}.$$

If $K = 3$, and if we similarly apply $\tau_{J_3}^{(B)}$ along the 3^{rd} dimension of the 3-dimensional array: $\tilde{c}_{(1:2J_1), (1:J_2), \begin{smallmatrix} 0 \\ 0 \end{smallmatrix}}^{0, 0, (1:J_3)}(z, \underline{\mathbf{J}})$, we will get 3-dimensional array:

$$\tilde{c}_{(1:2J_1), (1:J_2), (1:J_3)}^{0, 0, 0}(z, \underline{\mathbf{J}}) = \{\{\tilde{c}_{j_1, j_2, j_3}^{0, 0, 0}(z, \underline{\mathbf{J}})\}\}_{\{\underline{\mathbf{j}} \in \mathbb{N}_{(\underline{\mathbf{J}})}\}}.$$

We also realize that, if $K > 3$ and if we recursively keep on applying the linear transformation $\tau_{J_k}^{(B)}$ along k^{th} dimension of the output arrays for $k = 4, \dots, K$, after $K - 3$ such transformations, we will end up the the K -dimensional array:

$$\tilde{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}}) = \{\{\tilde{c}_{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}})\}\}_{\{\mathbf{j} \in \mathbb{N}_{\{\underline{\mathbf{J}}\}}\}}.$$

Now, let us fix the indices $j_3, \dots, j_K$ and traverse through the 1^{st} and 2^{nd} dimensions of this K -dimensional array: $\tilde{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$. We simultaneously change the first index j_1 over the values: $j_1 = 1, 2, \dots, 2J_1$ along with the second index j_2 over: $j_2 = 1, 2, \dots, J_2$, while we keep $j_3, \dots, j_K$ fixed and we look for every term $\tilde{c}_{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}})$ for such combinations of indices in $\mathbb{N}_{\{\underline{\mathbf{J}}\}}$, thus, we get the matrix:

$$\begin{aligned} \tilde{c}_{(1:2J_1), (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}}) &= \{\{\tilde{c}_{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}})\}\}_{1 \leq j_1 \leq 2J_1, 1 \leq j_2 \leq J_2} \\ &= \{\{c_{j_1}(z) \cdot \dot{c}_{j_2}(z, J_1) \cdot \prod_{k=3}^K \dot{c}_{j_k}(z, L_{k-1})\}\}_{1 \leq j_1 \leq 2J_1, 1 \leq j_2 \leq J_2} \\ &= \{\{c_{j_1}(z) \cdot \dot{c}_{j_2}(z, J_1)\}\}_{1 \leq j_1 \leq 2J_1, 1 \leq j_2 \leq J_2} \cdot \prod_{k=3}^K \dot{c}_{j_k}(z, L_{k-1}) \\ &= \tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2) \cdot \prod_{k=3}^K \dot{c}_{j_k}(z, L_{k-1}). \end{aligned}$$

Then, we realize that $\tilde{c}_{(1:2J_1), (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}})$ is a scalar-multiple of the matrix: $\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1)$ with multiplier $\prod_{k=3}^K \dot{c}_{j_k}(z, L_{k-1})$. Then, for $\tau_{J_1, J_2}^{(C)}$ as in `chev:lem:tr2Dto1D` we get:

$$\begin{aligned} \tau_{J_1, J_2}^{(C)}(\tilde{c}_{(1:2J_1), (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}})) &= \tau_{J_1, J_2}^{(C)}(\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2) \cdot \prod_{k=3}^K \dot{c}_{j_k}(z, L_{k-1})) \\ &= \tau_{J_1, J_2}^{(C)}(\tilde{c}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)) \cdot \prod_{k=3}^K \dot{c}_{j_k}(z, L_{k-1}) \\ &= c_{(1:2J_1 J_2)}(z) \cdot \prod_{k=3}^K \dot{c}_{j_k}(z, L_{k-1}) \\ &= \tilde{c}_{(1:2J_1 J_2), j_3, \dots, j_K}(z, \underline{\mathbf{J}}_{[K-1]}) \\ &= (\tilde{c}_{1, j_3, \dots, j_K}(z, \underline{\mathbf{J}}_{[K-1]}), \dots, \tilde{c}_{2J_1 J_2, j_3, \dots, j_K}(z, \underline{\mathbf{J}}_{[K-1]})); \\ &\text{where } \underline{\mathbf{J}}_{[K-1]} = (J_1 J_2, J_3, \dots, J_K) = (L_2, J_3, \dots, J_K) \in \mathbb{N}^{+K-1}; \text{ and} \end{aligned}$$

$$\tilde{c}_{j, j_3, \dots, j_K}(z, \underline{\mathbf{J}}_{[K-1]}) = c_j(z) \cdot \prod_{k=3}^K \dot{c}_{j_k}(z, L_{k-1})$$

for $j \in \{1, \dots, 2J_1 J_2\}$.

From our notation, we have:

$$\mathbb{N}_{\{\underline{\mathbf{J}}_{[K-1]}\}} = \{1, \dots, 2J_1 J_2\} \times \dots \times \{1, \dots, J_K\} \subset \mathbb{N}^{+K-1}$$

A general $\mathbf{j} = (j, j_3, \dots, j_K) \in \mathbb{N}_{\{\underline{\mathbf{J}}_{[K-1]}\}}$ is a $K-1$ dimensional integer vector. Hence, if we apply the linear transformation $\tau_{J_1, J_2}^{(C)}$ along 1^{st} and 2^{nd} dimensions of $\tilde{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$, while keeping the other indices fixed we will get the $K-1$ -dimensional array:

$$\tilde{c}_{(1:2J_1J_2), (1:J_3), \dots, (1:J_K)}(z, \underline{\mathbf{J}}_{[K-1]}) = \{\{\tilde{c}_{\mathbf{j}}(z, \underline{\mathbf{J}}_{[K-1]})\}\}_{\{\mathbf{j} \in \mathbb{N}_{\{\underline{\mathbf{J}}_{[K-1]}\}}\}}.$$

If $K = 3$, at this stage we have the matrix valued statistic $\tilde{c}_{(1:2J_1J_2), (1:J_3)}(z, \underline{\mathbf{J}}_{[2]})$, where $\underline{\mathbf{J}}_{[2]} = (J_1J_2, J_3)$. If we similarly apply the linear transformation $\tau_{J_1J_2, J_3}^{(C)}$ along 1^{st} and 2^{nd} dimensions of $\tilde{c}_{(1:2J_1J_2), (1:J_3)}(z, \underline{\mathbf{J}}_{[2]})$, we will get the array of length $J = \prod_{k=1}^3 J_k$, which is: $c_{(1:2J)}(z)$. We also realize, for $K > 3$ if we recursively keep on applying the linear transformation $\tau_{L_k, J_{k+1}}^{(C)}$ along 1^{st} and 2^{nd} dimensions of the output arrays, after $K-3$ such transformations, we will finally end up with the array of length $J = \prod_{k=1}^K J_k$, which is: $c_{(1:2J)}(z)$. So we have demonstrated a sequence of linear transformation that transforms the K -dimensional array $\tilde{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$ into the array $c_{(1:2J)}(z)$. In other words we have established the existence of a linear transformation $\tau_{\underline{\mathbf{J}}}^{(K)}$ so that:

$$c_{(1:2J)}(z) = \tau_{\underline{\mathbf{J}}}^{(K)}(\tilde{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})).$$

Remember, we defined $c_0(z) = 1$, let us define:

$$c_{(0:2J)}(z) = (c_0(z), c_1(z), \dots, c_{2J}(z)) = (1, c_{(1:2J)}(z)).$$

$c_{(0:2J)}(z)$ is indexed by the index set $\mathbb{N}_J = \{0, 1, \dots, 2J\}$. Now remember, for $\underline{\mathbf{J}} = (J_1, \dots, J_K)$, we defined the index set:

$$\mathbb{N}_{\{\underline{\mathbf{J}}\}} = \{1, \dots, 2J_1\} \times \{1, \dots, J_2\} \times \dots \times \{1, \dots, J_K\}.$$

We add the element 0 to this index set to define a new index set:

$$\mathbb{N}_{\underline{\mathbf{J}}} = \{0\} \cup (\{1, \dots, 2J_1\} \times \{1, \dots, J_2\} \times \dots \times \{1, \dots, J_K\}).$$

Also, for the new index element 0, define: $\tilde{c}^0(z, \underline{\mathbf{J}}) = 1$. Now we define:

$$\tilde{c}^0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}}) = \{\{\tilde{c}^{\mathbf{j}}(z, \underline{\mathbf{J}})\}\}_{\{\mathbf{j} \in \mathbb{N}_{\underline{\mathbf{J}}}\}}$$

Then, we have the following lemma:

Lemma 6 For $K \in \mathbb{N}^+$ and $\underline{\mathbf{J}} \in \mathbb{N}^{+K}$, $\exists$ a linear-transformation

$$\mathcal{T}_{\underline{\mathbf{J}}}^{(K)} : \mathbb{R}^{2J_1} \times \dots \times \mathbb{R}^{J_K} \rightarrow \mathbb{R}^{2J} \text{ such that:}$$

$$c_{(0:2J)}(z) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)}(\tilde{c}^0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})).$$

Proof 6 If $K = 1$ then $\underline{\mathbf{J}} = (J_1)$ and $J = J_1$. Take $\mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\mathbf{v}) = \tau_{J_1}^{(A)}(\mathbf{v})$, here as $\tau_{J_1}^{(A)}$ is same as in *chev:lem:trA* (for $J = J_1$). Then we have:

$$c_{(1:2J)}(z) = c_{(1:2J_1)}(z) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(c_{(1:2J_1)}(x)) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\tilde{c}_{(1:2J_1)}(z, \underline{\mathbf{J}})).$$

If $K = 2$ then $\underline{\mathbf{J}} = (J_1, J_2)$ and $J = J_1 J_2$. Take $\mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\mathbf{v}) = \tau_{J_1, J_2}^{(C)}(\tau_{J_1, J_2}^{(A, B)}(\mathbf{v}))$, here $\tau_{J_1, J_2}^{(A, B)}$ is same as in `chev:lem:tr2Dto2D` and $\tau_{J_1, J_2}^{(C)}$ is same as in `chev:lem:tr2Dto1D`. Then we have:

$$\begin{aligned} \mathbf{c}_{(1:2J)}(z) &= \tau_{J_1, J_2}^{(C)}(\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)) \\ &= \tau_{J_1, J_2}^{(C)}\left(\tau_{J_1, J_2}^{(A, B)}(\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2))\right) \\ &\Leftrightarrow \mathbf{c}_{(1:2J)}(z) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, \underline{\mathbf{J}})). \end{aligned}$$

If $K > 2$ then $\underline{\mathbf{J}} = (J_1, J_2, \dots, J_K)$ and $J = \prod_{k=1}^K J_k$. Take $\mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\mathbf{v}) = \tau_{\underline{\mathbf{J}}}^{(K)}(\mathbf{v})$, here $\tau_{\underline{\mathbf{J}}}^{(K)}$ is same as in `chev:lem:trKDto1D`

$$\mathbf{c}_{(1:2J)}(z) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\mathbf{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})).$$

Then we have the generic linear-transformation $\mathcal{T}_{\underline{\mathbf{J}}}^{(K)'} : \mathbb{R}^{2J_1} \times \dots \times \mathbb{R}^{J_K} \rightarrow \mathbb{R}^{2J}$ for $K \in \mathbb{N}^+$ and $\underline{\mathbf{J}} \in \mathbb{N}^{+K}$ satisfying:

$$\mathbf{c}_{(1:2J)}(x) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\mathbf{c}_{(1:2J_1), \dots, (1:J_K)}(x, \underline{\mathbf{J}})).$$

Now, observe that for $K \in \mathbb{N}^+$ and $\underline{\mathbf{J}} \in \mathbb{N}^{+K}$, we can break down the generic transformation $\mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}$, into small simple components: $\tau_{\underline{\mathbf{J}}}^{(A)}$, $\tau_{\underline{\mathbf{J}}}^{(B)}$, $\tau_{J_1, J_2}^{(A, B)}$ and $\tau_{J_1, J_2}^{(C)}$. All of these simple transformation gives a linear-combination of the input vector as the output. Suppose $\mathcal{V}$ is the generic invertible transformation that vectorizes a multi-dimensional array in the natural-order of the index-set of the array. Then, we will be able to find a $2J \times 2J$ matrix $\mathcal{A}_{\underline{\mathbf{J}}}^{(K)'}$, such that:

$$\mathbf{c}_{(1:2J)}(x) = \mathcal{A}_{\underline{\mathbf{J}}}^{(K)'} \cdot \mathcal{V}(\mathbf{c}_{(1:2J_1), \dots, (1:J_K)}(x, \underline{\mathbf{J}})).$$

Now observe that:

$$\mathcal{V}(\mathbf{c}_{(0:2J_1), \dots, (1:J_K)}(x, \underline{\mathbf{J}})) = \begin{pmatrix} 1 \\ \mathcal{V}(\mathbf{c}_{(1:2J_1), \dots, (1:J_K)}(x, \underline{\mathbf{J}})) \end{pmatrix}$$

Take $\mathcal{A}_{\underline{\mathbf{J}}}^{(K)'} := \text{diag}(1, \mathcal{A}_{\underline{\mathbf{J}}}^{(K)'})$, since $\mathbf{c}_{(0:2J)}(z) = (1, \mathbf{c}_{(1:2J)}(z))$, we must have

$$\mathbf{c}_{(0:2J)}(x) = \mathcal{A}_{\underline{\mathbf{J}}}^{(K)} \cdot \mathcal{V}(\mathbf{c}_{(0:2J_1), \dots, (1:J_K)}(x, \underline{\mathbf{J}})).$$

So, we have proved the existence of a transformation $\mathcal{T}_{\underline{\mathbf{J}}}^{(K)}$, such that:

$$\mathbf{c}_{(0:2J)}(x) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)}(\mathbf{c}_{(0:2J_1), \dots, (1:J_K)}(x, \underline{\mathbf{J}})).$$

Now consider a p -dimensional vector $\mathbf{x} \in (0, 1)$, so that if $\mathbf{x} = (x_1, \dots, x_p)$, then $x_l \in (0, 1)$ for $1 \leq l \leq p$. Let us define the p -dimensional function:

$$\mathbf{c}_{(0:2J); p}(\mathbf{x}) = \otimes_{l=1}^p \mathbf{c}_{(0:2J)}(x_l).$$

Note that $\mathbf{c}_{(0:2J); p}(\mathbf{x})$ can be identified as a p -dimensional array with dimensions $(2J+1) \times \dots \times (2J+1)$ (p times). This p -dimensional array is indexed by the index set:

$$\mathbb{N}_{J; p} = \{0, \dots, 2J\} \times \dots \times \{0, \dots, 2J\} \text{ (} p \text{ times)}.$$

For $\mathbf{j} = (j_1, \dots, j_p) \in \mathbb{N}_{J;p}$, a general $\mathbf{j}^{th}$ element of $c_{(0:2J);p}(\mathbf{x})$ is the function:

$$c_{\mathbf{j}}(\mathbf{x}) = \prod_{l=1}^p c_{j_l}(x_l).$$

Let us also define:

$$\tilde{c}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{x}, \underline{\mathbf{J}}) = \otimes_{l=1}^p \left(\tilde{c}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(x_l, \underline{\mathbf{J}}) \right).$$

This array-like function is indexed by the index set: $\mathbb{N}_{\underline{\mathbf{J}};p}$, defined as:

$$\mathbb{N}_{\underline{\mathbf{J}};p} = \mathbb{N}_{\underline{\mathbf{J}}} \times \dots \times \mathbb{N}_{\underline{\mathbf{J}}} \text{ (} p \text{ times)}.$$

Let us denote a general index element of $\mathbb{N}_{\{\underline{\mathbf{J}}, p\}}$ as $\underline{\mathbf{j}}$, which is a p -tuple: $(\underline{\mathbf{j}}_1, \dots, \underline{\mathbf{j}}_p)$. For $1 \leq l \leq p$, $\underline{\mathbf{j}}_l$ is either 0 or a K -element-vector $(j_{l,1}, \dots, j_{l,K})$

Now, the $\underline{\mathbf{j}}^{th}$ element of $\tilde{c}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{x}, \underline{\mathbf{J}})$ is:

$$\tilde{c}^{\underline{\mathbf{j}}}(\mathbf{x}, \underline{\mathbf{J}}) = \prod_{l=1}^p \left(c^{j_{l,1}}(x_l) \cdot \prod_{k=2}^K \dot{c}^{j_{l,k}}(x_l, L_{k-1}) \right)^{1(\underline{\mathbf{j}}_l \neq 0)}.$$

We have the following lemma:

Lemma 7 *Let $\mathbf{x}$ is an element of the p -dimensional open interval $(0, 1)$, for $K \in \mathbb{N}^+$ and $\underline{\mathbf{J}} \in \mathbb{N}^{+K}$, $\exists$ a linear-transformation $\mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)}$, such that:*

$$c_{(0:2J)^P}(z) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)}(\tilde{c}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(z, \underline{\mathbf{J}})).$$

Proof 7 *From chev:lem:gen, we have for $1 \leq l \leq p$:*

$$c_{(0:2J)}(x_l) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)}(\tilde{c}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(x_l, \underline{\mathbf{J}})).$$

If we take tensor-product of both sides for $1 \leq l \leq p$, we get:

$$\begin{aligned} \otimes_{l=1}^p c_{(0:2J)}(x_l) &= \otimes_{l=1}^p \left(\mathcal{T}_{\underline{\mathbf{J}}}^{(K)}(\tilde{c}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(x_l, \underline{\mathbf{J}})) \right) \\ \Leftrightarrow c_{(0:2J)^P}(\mathbf{x}) &= \left(\otimes_{l=1}^p \mathcal{T}_{\underline{\mathbf{J}}}^{(K)} \right) \left(\otimes_{l=1}^p \tilde{c}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(x_l, \underline{\mathbf{J}}) \right) \\ \Leftrightarrow c_{(0:2J)^P}(\mathbf{x}) &= \mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)}(\tilde{c}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{x}, \underline{\mathbf{J}})). \end{aligned}$$

If we let $\mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)} := \otimes_{l=1}^p \mathcal{T}_{\underline{\mathbf{J}}}^{(K)}$, we have:

$$c_{(0:2J)^P}(\mathbf{x}) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)}(\tilde{c}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{x}, \underline{\mathbf{J}})).$$

1.5. Main theorem

Now remember, we have the following SEP statistic and standardized WEP statistic for $\mathbf{j} \in \mathbb{N}_{J;p}$:

$$C_{\mathbf{j}}(\mathbf{X}) = \sum_{i \in I} c_{\mathbf{j}}(\mathbf{x}_i), \quad \bar{C}_{\mathbf{j}}(\mathbf{X}) = (1/X) \cdot C_{\mathbf{j}}(\mathbf{X}).$$

Let us define the collections:

$$\begin{aligned} \mathbf{C}_{(0:2J)^P}(\mathbf{X}) &= \{\{\mathbf{C}_{\mathbf{j}}(\mathbf{X})\}\}_{\mathbf{j} \in \mathbb{N}_{J;P}} \\ \bar{\mathbf{C}}_{(0:2J)^P}(\mathbf{X}) &= \{\{\bar{\mathbf{C}}_{\mathbf{j}}(\mathbf{X})\}\}_{\mathbf{j} \in \mathbb{N}_{J;P}}. \end{aligned}$$

Note that:

$$\mathbf{C}_{(0:2J)^P}(\mathbf{X}) = \sum_{i \in I} \mathbf{c}_{(0:2J)^P}(\mathbf{x}_i), \quad \bar{\mathbf{C}}_{(0:2J)^P}(\mathbf{X}) = (1/\mathbf{x}) \cdot \mathbf{C}_{(0:2J)^P}(\mathbf{X}).$$

We define the following SEP statistic and standardized WEP statistic for a general $\underline{\mathbf{j}} \in \mathbb{N}_{\underline{\mathbf{J}};P}$:

$$\tilde{\mathbf{C}}^{\underline{\mathbf{j}}}(\mathbf{X}, \mathbf{J}) = \sum_{i \in I} \tilde{\mathbf{c}}^{\underline{\mathbf{j}}}(\mathbf{x}_i, \mathbf{J}), \quad \bar{\tilde{\mathbf{C}}}^{\underline{\mathbf{j}}}(\mathbf{X}, \mathbf{J}) = (1/\mathbf{x}) \cdot \tilde{\mathbf{C}}^{\underline{\mathbf{j}}}(\mathbf{X}, \mathbf{J}).$$

Again, we define the collections of statistics:

$$\begin{aligned} \tilde{\mathbf{C}}_{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{X}, \underline{\mathbf{J}}) &= \{\{\tilde{\mathbf{C}}^{\underline{\mathbf{j}}}(\mathbf{X}, \mathbf{J})\}\}_{\underline{\mathbf{j}} \in \mathbb{N}_{\underline{\mathbf{J}};P}} \\ \bar{\tilde{\mathbf{C}}}_{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{X}, \underline{\mathbf{J}}) &= \{\{\bar{\tilde{\mathbf{C}}}^{\underline{\mathbf{j}}}(\mathbf{X}, \mathbf{J})\}\}_{\underline{\mathbf{j}} \in \mathbb{N}_{\underline{\mathbf{J}};P}}. \end{aligned}$$

And, we have the identities:

$$\begin{aligned} \tilde{\mathbf{C}}_{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{X}, \underline{\mathbf{J}}) &= \sum_{i \in I} \tilde{\mathbf{c}}_{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{x}_i, \underline{\mathbf{J}}) \\ \bar{\tilde{\mathbf{C}}}_{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{X}, \underline{\mathbf{J}}) &= (1/\mathbf{x}) \cdot \tilde{\mathbf{C}}_{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{X}, \underline{\mathbf{J}}). \end{aligned}$$

Now, we are in a position to prove the main-result of this supplementary text: For $K \in \mathbb{N}^+$ and $\underline{\mathbf{J}} \in \mathbb{N}^{+K}$, $\exists$ a linear-transformation $\mathcal{T}_{\underline{\mathbf{J}}}^{(K,P)}$, such that:

$$\bar{\mathbf{C}}_{(0:2J)^P}(X) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K,P)}\left(\bar{\tilde{\mathbf{C}}}_{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(X, \underline{\mathbf{J}})\right).$$

Proof 8 From *chev:lem:gen:pd*, we have for $i \in I$:

$$\mathbf{c}_{(0:2J)^P}(\mathbf{x}_i) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K,P)}\left(\tilde{\mathbf{c}}_{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{x}_i, \underline{\mathbf{J}})\right).$$

Since $\mathcal{T}_{\underline{\mathbf{J}}}^{(K,P)}$ is a linear-transformation, if we take a sum over $i \in I$ and divide both sides by $|\mathbf{X}|$, we get:

$$\begin{aligned} \frac{\sum_{i \in I} \mathbf{c}_{(0:2J)^P}(\mathbf{x}_i)}{|\mathbf{X}|} &= \frac{\sum_{i \in I} \mathcal{T}_{\underline{\mathbf{J}}}^{(K,P)}\left(\tilde{\mathbf{c}}_{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{x}_i, \underline{\mathbf{J}})\right)}{|\mathbf{X}|} \\ \Leftrightarrow \frac{\sum_{i \in I} \mathbf{c}_{(0:2J)^P}(\mathbf{x}_i)}{|\mathbf{X}|} &= \mathcal{T}_{\underline{\mathbf{J}}}^{(K,P)}\left(\frac{\sum_{i \in I} \tilde{\mathbf{c}}_{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(\mathbf{x}_i, \underline{\mathbf{J}})}{|\mathbf{X}|}\right) \\ \Leftrightarrow \bar{\mathbf{C}}_{(0:2J)^P}(X) &= \mathcal{T}_{\underline{\mathbf{J}}}^{(K,P)}\left(\bar{\tilde{\mathbf{C}}}_{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^P}(X, \underline{\mathbf{J}})\right). \end{aligned}$$

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